

DP Computer Science: B 1.1.1 Problem Specification

1. Core Concepts & Learning Objectives

Learning Intentions

By the end of this lesson, students will be able to:

- Define and explain the key elements of a problem specification:
 - Problem statement
 - Constraints and limitations
 - Objectives and goals
 - Input specifications
 - Output specifications
 - Evaluation criteria
 - Identify and articulate the key elements of a problem specification given a **real-world scenario**
 - Construct a **complete and well-defined problem specification** for a given problem
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Key Vocabulary

Term	Definition
Problem specification	A blueprint for solving a problem; defines the problem, constraints, goals, inputs, outputs, and evaluation criteria
Problem statement	A clear and concise description of the problem to be solved
Constraints	Limitations or restrictions that must be considered (e.g., time, resources, rules)
Goals	The desired outcome or objective of the solution
Input specifications	The data that is provided to the system

Term	Definition
Output specifications	The result expected from the system
Evaluation criteria	The standards or measures used to assess the success of a solution

ATL Skills

Skill	Description
Thinking (Critical Thinking)	Analyse information, identify problems, and evaluate solutions
Thinking (Creative Thinking)	Generate innovative solutions and explore different approaches to problem-solving
Information Literacy	Access, evaluate, and use information effectively
Communication	Clearly and concisely articulate ideas and solutions
Self-Management	Plan and manage their work effectively

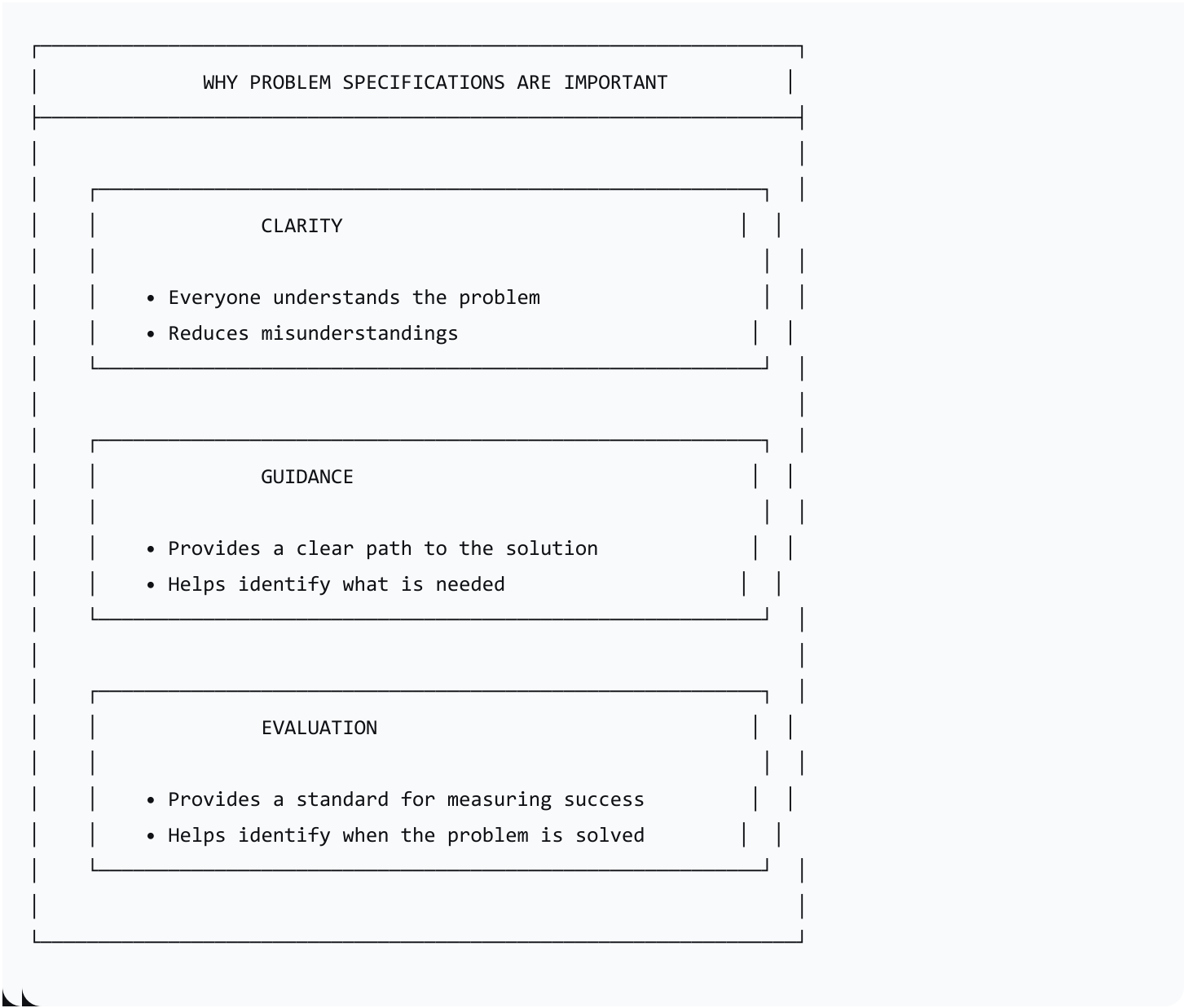
2. Introduction to Problem Specifications

The Blueprint Analogy

A problem specification is like a blueprint for building a house—it defines what needs to be built, what materials can be used, and what the final result should look like.

Why Do We Need Problem Specifications?

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The Cake Baking Analogy

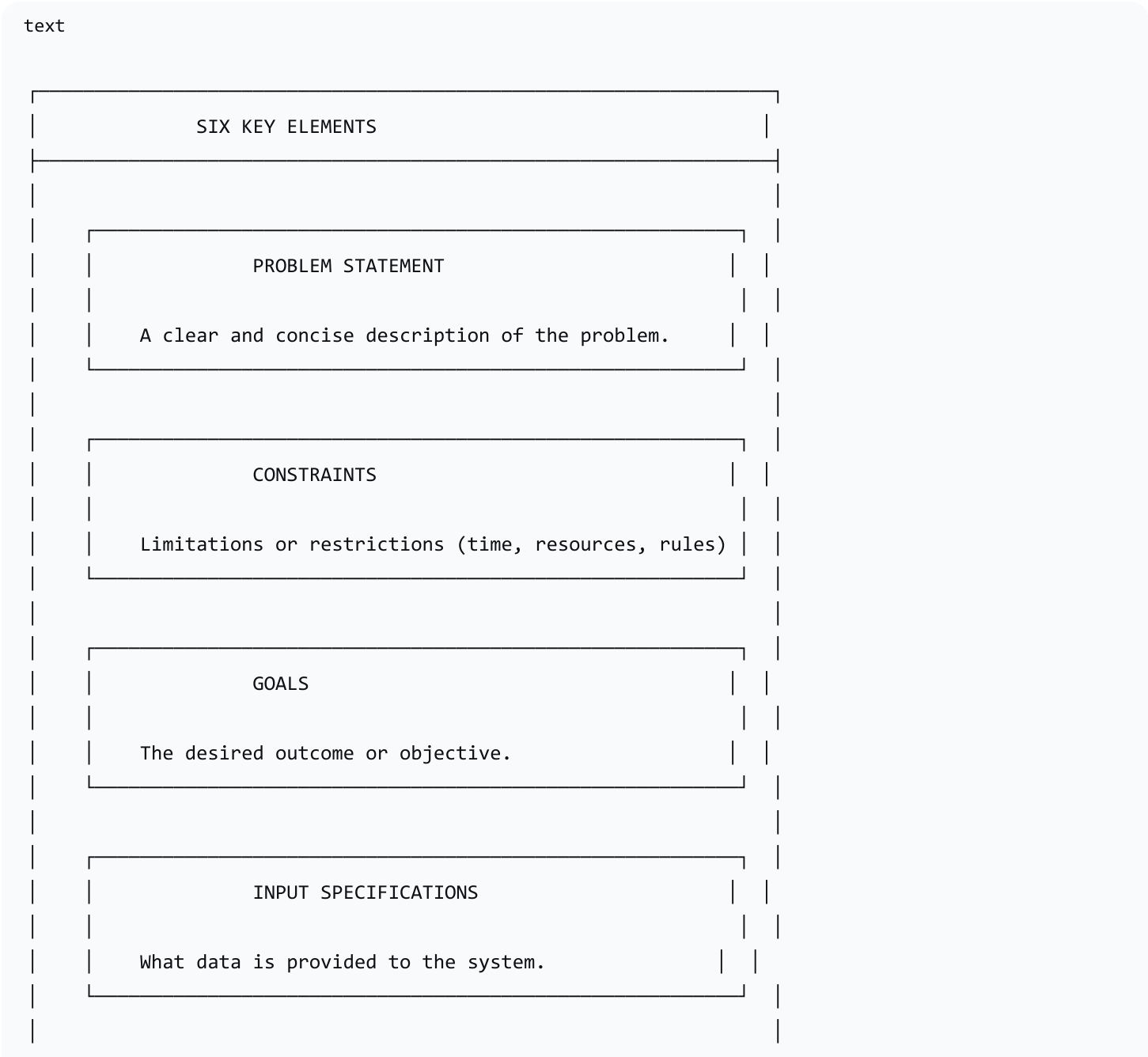
"Imagine you need to bake a cake for a friend's birthday. What do you need to consider before you start?"

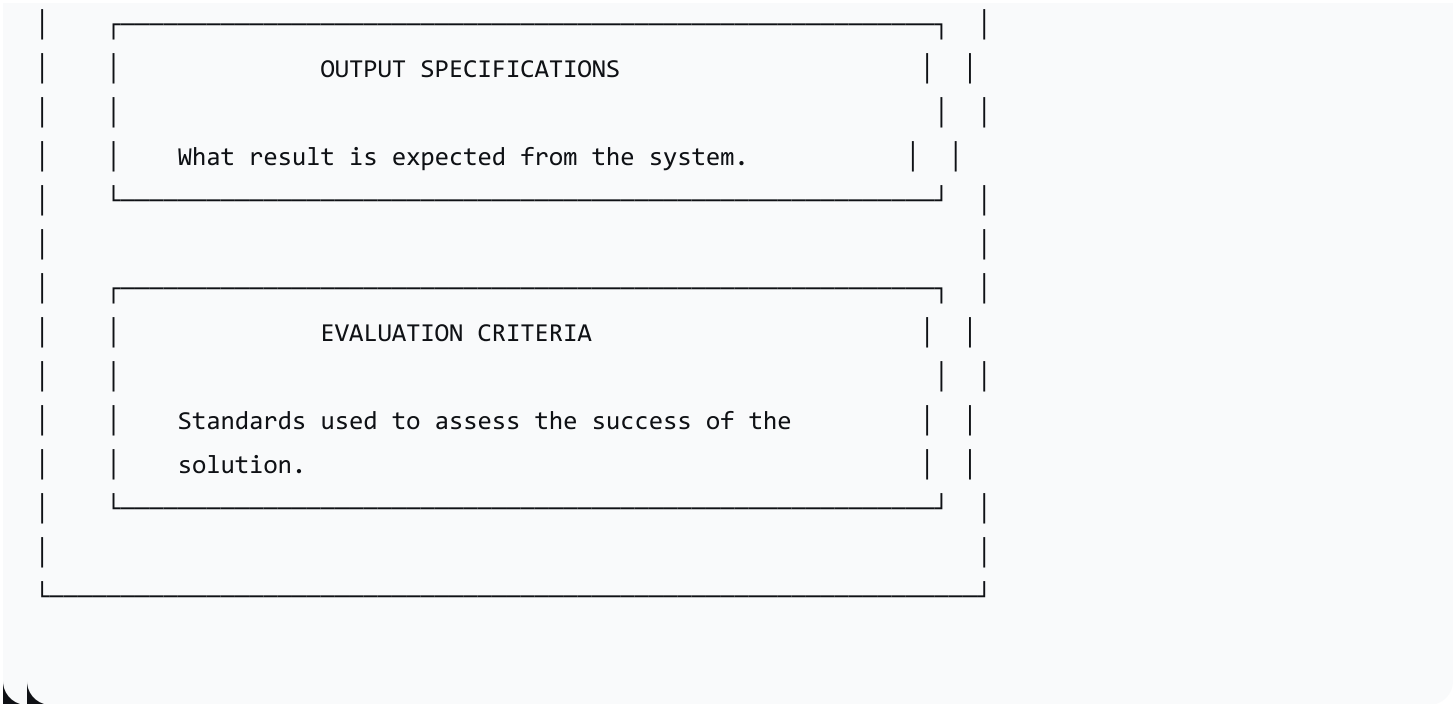
Consideration	Problem Specification Element
"What type of cake?"	Problem Statement
"What ingredients do I have?"	Constraints
"What should the cake taste like?"	Goals
"What ingredients do I need?"	Input Specifications

Consideration	Problem Specification Element
"What does the finished cake look like?"	Output Specifications
"How will I know if it's successful?"	Evaluation Criteria

3. Key Elements of a Problem Specification

The Six Key Elements





4. Worked Example: Vending Machine

Problem Scenario

Design a program for a vending machine that dispenses snacks.

Problem Specification

Element	Description
Problem Statement	Design a program for a vending machine that dispenses snacks when the correct code and sufficient money are entered
Constraints	Limited options (5 snacks), accepts only specific coins, must provide change
Goals	Dispense the correct snack when the corresponding code and sufficient money are entered
Input Specifications	Snack code, money

Element	Description
Output Specifications	Snack, change (if applicable)
Evaluation Criteria	Accuracy, Efficiency, Reliability, Usability, Maintainability

Evaluation Criteria Breakdown

Criterion	Description
Accuracy	Does the machine dispense the correct snack and change every time?
Efficiency	Does the program calculate change quickly and accurately?
Reliability	Does the program handle invalid inputs (incorrect codes, invalid coins) and unexpected situations (e.g., coin jams)?
Usability	Is the user interface easy to understand and use?
Maintainability	Is the code well-organised and easy to understand for future modifications?

5. Practice Scenario: Library Book Search

Problem Scenario

| Create a program to help users find books in a library database.

Task: Identify the Key Elements

Element	Your Answer
Problem Statement	
Constraints	

Element	Your Answer
Goals	
Input Specifications	
Output Specifications	
Evaluation Criteria	

Suggested Answer

Element	Description
Problem Statement	Create a program to help users find books in a library database
Constraints	Database has a specific format; search must be by title, author, or keyword
Goals	Display a list of relevant books with their location in the library
Input Specifications	Search query (title, author, keyword)
Output Specifications	List of matching books
Evaluation Criteria	Accuracy, Completeness, Efficiency, Usability, Relevance

6. Practice Scenario: Online Quiz

Problem Scenario

| Develop an online quiz with multiple-choice questions.

Suggested Answer

Element	Description
Problem Statement	Develop an online quiz with multiple-choice questions
Constraints	Time limit for the quiz; questions must be randomly selected from a pool
Goals	Present questions, record answers, calculate and display the final score
Input Specifications	User's answers to questions
Output Specifications	Quiz score, feedback
Evaluation Criteria	Accuracy, Randomisation, Security, Usability, Feedback

7. Practice Scenario: Social Media Scheduler

Problem Scenario

| Create a program that allows users to schedule posts on different social media platforms.

Suggested Answer

Element	Description
Problem Statement	Create a program that allows users to schedule posts on different social media platforms
Constraints	Must connect to different social media APIs; handle different post formats; comply with platform terms of service
Goals	Allow users to schedule posts in advance; automatically publish posts at scheduled times; track post performance

Element	Description
Input Specifications	Social media account credentials; post content; scheduled date and time; platform selection
Output Specifications	Confirmation of scheduled posts; notifications of successful posting; reports on post-performance
Evaluation Criteria	Reliability, Compatibility, Usability, Performance Tracking, Security, Efficiency

8. Group Activity: Analysing Problem Scenarios

Instructions

- 1. Divide into small groups
- 2. Each group receives a problem scenario
- 3. Identify and write down the key elements of each problem specification
- 4. Discuss and collaborate within your group
- 5. Share your findings with the class

Scenarios

Scenario	Description
Scenario 1	Vending Machine
Scenario 2	Library Book Search
Scenario 3	Online Quiz
Scenario 4	Social Media Scheduler

9. Lesson Sequence

Lesson Plan (60 minutes)

Phase	Time	Activity
Introduction	10 min	Teacher-led discussion; baking a cake analogy; introduce problem specification
Key Elements	15 min	Define and explain each element with examples
Group Activity	20 min	Students analyse problem scenarios and identify key elements
Class Discussion	10 min	Groups share findings; compare and contrast; address misconceptions
Conclusion	5 min	Recap; eBook quiz; real-world application

10. Assessment Activities

Assessment Type	Description
Class Participation	Observe engagement and contributions
Group Activity	Assess ability to identify key elements of a problem specification
Quiz	eBook quiz on problem specification elements
Exit Ticket	Gauge understanding and identify areas needing clarification

Sample Exit Ticket Questions:

1. What is a problem specification?
2. Name three key elements of a problem specification.
3. What is the purpose of evaluation criteria?
4. Why is it important to define constraints when solving a problem?

11. Differentiation

Level	Strategy
Support	Provide sentence starters; use relatable examples; work in groups
Challenge	Create a problem specification for a complex, real-world scenario

12. Resources

- Presentation (Slide Deck)
- eBook (B1.1.1 with examples)
- Quizlet (B1.1.1 vocabulary flashcards)
- Activity: Problem scenario analysis

13. Summary: Key Takeaways

Element	Key Point
Problem Statement	What is the problem?
Constraints	What are the limits?
Goals	What should the solution achieve?
Input Specifications	What data is provided?
Output Specifications	What result is expected?
Evaluation Criteria	How will success be measured?

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KEY REMINDERS

- ✓ Problem specification = blueprint for solving problems
- ✓ Defines the problem, constraints, goals, inputs, outputs, and evaluation criteria
- ✓ Provides clarity, guidance, and a standard for evaluation
- ✓ Essential for effective problem-solving

"A well-defined problem specification is the foundation of an effective solution."